

Affective intelligence — the missing link?

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Scientists have recently learned of many surprising roles played by human emotion — especially in cognitive processes such as perception, decision making, memory, judgement, and more. Moreover, psychologists have redefined human intelligence to include emotional intelligence, especially the ability to accurately recognise and express affective information. This paper suggests that ‘affective intelligence’, the communication and management of affective information in human/computer interaction, is a key link that is missing in telepresence environments and other technologies that mediate human/human communication. This paper discusses new research in affective intelligence, and how it can impact upon and enhance the communication process, allowing the delivery of the more natural interaction that is critical for a true sense of telepresence.

1. Introduction

The topic of emotion is one that many engineers and scientists would prefer to eschew. After all, emotional behaviour is regarded as irrational — deficient in reasoning — and counter to the entire scientific method. Too much emotion, poorly managed, is clearly undesirable. However, there is another side to emotion that is only recently beginning to be understood, a side that is essential for rational human behaviour. Scientists have discovered that not only is too much emotion detrimental to decision making and rational behaviour, but so too is too little emotion. It is time to take a closer look at the communication and management of emotion in human/computer interaction, what is here called ‘affective intelligence’, and to consider how this will have an impact on developments in technology and machine intelligence.

In this paper, the words ‘emotional’ and ‘affective’ are used interchangeably as adjectives describing both physical and cognitive aspects of emotion, although ‘affective’ will sometimes be used in a broader sense than ‘emotional’, particularly when including computers and other communications technologies. Note that the word ‘feeling’ does not always imply emotion; squeezing scrambled eggs between your fingers feels squishy, and is a tactile feeling, not an emotional one. Feelings such as hunger and pain are also usually not considered emotion. On the other hand, having a good feeling or a bad feeling about something — so-called ‘gut feelings’ — are known to be important valenced emotional indicators.

Many years ago, Gardner revolutionised thinking about human intelligence with his theory of ‘multiple intelligences’ [1] which includes not only the traditional mathematical and verbal abilities, but also other abilities, especially social intelligence, which consists of interpersonal and intrapersonal skills. Later, Salovey and

Mayer identified these skills as ‘emotional intelligence’ which they define as ‘the ability to monitor one’s own and others’ feelings and emotions, to discriminate between them and to use this information to guide one’s thinking and actions’ [2]. The importance of these skills has been underscored by Goleman in his book [3] which argues that affective abilities are more important than traditional IQ for predicting success in life.

Emotional intelligence in people involves many factors such as the ability to accurately recognise emotions in others, to be aware of them in oneself, to express them in an appropriate and controlled way, and to manage emotions, for self-motivation, persistence, and social deftness. But, what does all of this have to do with technology?

Today it is easy to find people who spend more time interacting with a computer than with other humans. Every minute people enter the on-line communities of the Internet where they communicate with each other through computers or other mediating technologies. Daily interaction between humans and computers has billions of dollars of economic impact, not to mention psychological impact, which is harder to quantify. Furthermore, when humans interact with technology, they do so in a very interesting way — naturally and socially, almost as if it was no different than interacting directly with a person.

The claim of social and natural interaction is based on the findings of Reeves and Nass at Stanford University, who have conducted dozens of classical tests of human social interaction by substituting computers into a role usually occupied by humans. Hence, a test that traditionally studies a human/human interaction was used to study a human/computer interaction. For example, one experiment

examined how what is said by human *A* about human *B*'s performance changes when *A* gives the evaluation face-to-face with *B*, as opposed to when *A* gives the evaluation about *B* to another (presumably neutral) person. Studies of human social interaction indicate that, in general, humans are nicer face-to-face. In a variation on the traditional test, human *B* is replaced with computer *B*. Human *A* now has to evaluate the computer's performance after the computer has given *A*, say, a short lesson. Human *A* gives *B* its evaluation 'face-to-face', and then is asked by a different computer for an evaluation of how *B* did. The classic human/human results still held; for example, the tendency to be nicer face-to-face remained. Numerous similar experiments were done by Reeves and Nass, revealing that the classic results of human/human studies were maintained in human/computer studies. The findings held true even for people who 'know better', such as computer science students who know that computers do not have emotions. After accounting for potential biasing factors, the researchers concluded that individuals' interactions with computers are inherently natural and social. For details of the experiments described here, see Nass, Steuer and Tauber [4], Nass and Sundar [5], and Reeves and Nass [6].

When you see a person expressing anger at their computer, this behaviour is an example of a natural and social interaction. They know that the computer does not recognise their anger, and that it does not have or understand anger, but they still express anger at it nonetheless. Emotion is a natural and social part of human/human communication, and people naturally use it when they interact with computers and other forms of media technology. Consequently, the critical role of emotional intelligence in human/human interaction applies to human/computer interaction, and to interaction between humans

which is mediated by computers — human/computer/human interaction (see Fig 1). The latter is particularly important in telepresence environments, where the critical factor in connecting geographically remote people is the natural link — human to human.

2. Communicating emotion — from e-mail to telepresence

Telecommunications is playing an ever-increasing role in today's society, within business, education and at home. It has developed from its historical past into being a medium that not only transports voice, but is now a critical business component. However, the communications interface has developed less rapidly and while video conferencing currently offers the richest form of telepresence today, examples of on-line communities [7] are now challenging this position. Stimulating our senses with experience or illusion is a key element in delivering a more natural communications environment. Turin, of London University, compares our sensory system to the most wonderful personal assistant — always there, beavering away for us, never complaining, and constantly underrated. Scientists working in artificial intelligence have found out just how amazing our perceptual abilities are in trying to replicate human perception with machines.

Stimulating this amazing machine has been a vision of leading writers such as Kubrick and Clarke for many years, and the ability to appear collocated in multiple locations or worlds, which these authors have implied, has become a major business driver for the telecommunications industry today. The quote 'Beam me up, Scottie' is probably familiar to all, but can one actually appear in an alternative location, with all of the benefits and feelings of a real-world

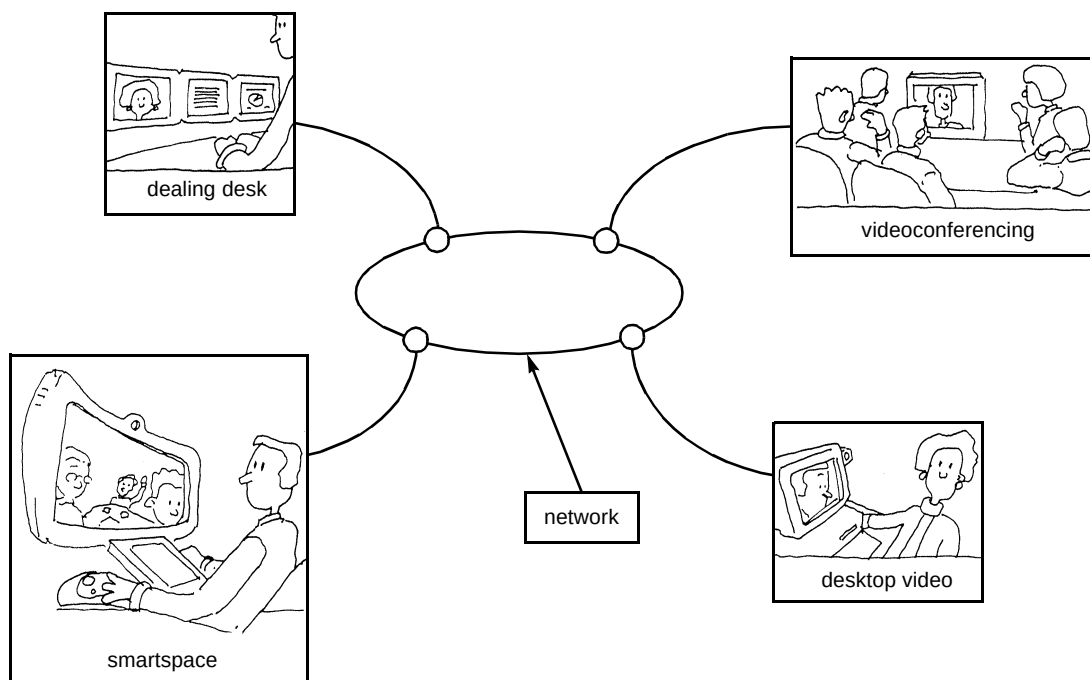


Fig 1 Human/human computer-mediated communication.

interaction? Videoconferencing has opened up the possibilities for this vision with face-to-face meetings now an everyday occurrence within the business sector. However, to date the quality of presence is somewhat limited and research shows that the interaction is poor if the participants have not previously met in person [8]. Research at BT Laboratories (BTL) has started to look into what differences are experienced between the real-world face-to-face encounter and the same situation over a networked connection. Preliminary results show that the feeling of presence is considerably reduced when communicated from a remote location [8]. Clearly this statement needs to be qualified in that the application, task, latency, cognitive load, intimacy of the participants and pre-conditioning will all affect the perception of the interaction.

Today's videoconferencing terminals are opening up new ways of working for many business applications. However, as business becomes more like a molecular structure where each node represents a virtual partner participating from the most appropriate corner of the globe, then high-quality eye-to-eye contact becomes increasingly critical to the new business environment. This will become increasingly important as virtual business partnerships develop in this way. High-quality videoconferencing conveys facial expression effectively, but it still fails to deliver a strong sense of presence. For example, we can detect the wink of an eye or squint, but may be totally unaware that this was due to a video camera light disturbing the participant. The shared experience of meeting face-to-face in the real world is somehow lost when conventional videoconferencing is used. Experience is more than just eye-to-eye contact, it relies on the assimilation of a wide range of human gestures and on building an appreciation of the surrounding physical environment. In particular, some of the key pieces that are missing are the physical handshake, and subtle cues of affective timing and expression. Affective communication is generally impaired by current forms of mediated communication, and when the bandwidth of affective communication is reduced, then problems can occur.

E-mail, for example, is one of the most affect-limited forms of communication. Very few people have the skill to masterfully communicate feelings using text only. Almost all people that rely on e-mail have wasted one or more days trying to straighten out a misunderstanding generated by a hastily produced message that offended the recipient. It was not intended as obnoxious, and had the same words been delivered as a casual spoken comment — even had they been said on the telephone — the misunderstanding may well have been avoided. Instead, the recipient detected a nasty tone in the text and took offence. E-mail is affect-limited, because it carries very little intonation or information about the affective state of the sender. Affect is so important to the communication process that when it is constrained people find some way, however limited, to add this facet to their communication. Actors and animators

exaggerate affective expression, because it is crucial that emotion be communicated accurately [9]. Users of e-mail have developed limited means for expressing their emotions. Symbol combinations like :-) and :- (are now in common use. However, these 'emoticons' are a woefully inadequate way to express the range of natural human emotions. Figure 2 shows how 'emotions' are being used in an on-line chat world. This strategy places a burden on the user that is unnatural.

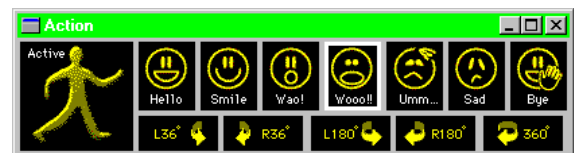


Fig 2 Emoticons.

It is interesting to note that although affect might not be communicated by a piece of technology, people still tend to perceive it — to fill it in when it is missing. Even a neutral message may be perceived as emotional. Studies have shown that the mood of the perceiver influences perception of ambiguous stimuli. If healthy subjects are asked to quickly jot words they hear, then they are more inclined to spell 'presents' than 'presence' if they are happy, and to spell 'banned' than 'band' if they are sad. Subjects resolve lexical ambiguity in homophones in a mood-congruent fashion [10]. Similar results occur when subjects look at ambiguous facial expressions. Depressed subjects judge the faces as having more rejection and sadness [11]. Moods also bias perception of the likelihood of events — an individual in a negative mood perceives negative events as more likely and positive events as less likely, and the reverse holds true for people in positive moods. A variety of studies has indicated that not only does affect play a critical role in perception and judgement, but it also plays a critical role in creativity and other aspects of cognition [12].

Affective intelligence is a new area of research, being explored by BT Laboratories and the MIT Media Lab, which focuses on understanding how emotion is communicated and managed in human/computer interaction, and in human/human interaction when that interaction is mediated by technology. In particular, affective intelligence involves communication of affect — recognising human affective expression, and transmitting it in a meaningful way. Examples are a synthetic avatar representing a real human and mimicking the expressive gestures and intonation, and a software agent that is trying to adapt to its user, and recognises its user's expression of dislike and like, in order to figure out when it needs to modify its behaviour. In these cases and others like them, the role of affect is a key link in facilitating communication. The focus of the present research is to establish some means of recognising affect, and to understand how this new information can contribute to more natural human/computer interfaces and telepresence applications. In so doing, the so-

called hidden elements of human communication can be treated scientifically, and the technology can be adjusted to overcome some of its present limitations.

Understanding and accommodating affective communication is a key part of affective intelligence. In particular, affective communication is usually subtle, and it can be important to preserve its subtlety. Consider, for example, the problem of trying to run a brainstorming meeting when all the participants are not physically in the same place. If you try to do this by e-mail, it will be easy to collect a lot of ideas, but it will be hard to 'sense' when one of the ideas appeals to everyone. This is critical, because people tend to build on the ideas that they sense people like. The way that 'liking' tends to be communicated is by subtle affective cues — did everyone perk up and look interested when that idea was spoken? Did people lean forward or raise their eyebrows? Did they respond quickly, or with hesitation? Did people slump back, exhale, or frown? These subtle affective cues cannot be replaced by more overt expressions, such as asking people to rank every idea as it is expressed, which would be unnatural and upset the creative process. However, these things are not communicated by e-mail, and only a few of them are communicated by present videoconferencing technology. The new enhanced reality environments being developed at BT [13, 14] aim to address these deficiencies.

Consider the application of affective intelligence to such new scalable collaborative environments. Traditional videoconferencing is not scalable; as more remote parties join the conference, each appearing to look out of the screen, it becomes increasingly difficult for participants to appreciate the shared experience. In scalable collaborative environments, participants share space in a 3-D virtual world, enabling communities to interact in a way that is more analogous to typical human interaction. However, the behaviours or rules of engagement within these shared environments need to be governed so as to mediate the interaction between the real-world surroundings within which the user is located and the computer-generated world within which they are engaged. For example, if a user is interrupted within your real-world space, then the affective interpretation of their presence within the shared environment will need to be addressed. If one person responds quickly, leaning forward with interest, then this needs to be communicated. Even subtle differences in timing — instantaneous versus hesitating — are important for judging affective response. These environments need enough affective intelligence to recognise what cues are important, no matter how subtle, and to preserve the communication of these cues.

The blurring of the real world with the computer-generated world is of particular importance in emerging communication environments. The concept of immersion is relevant with respect to traditional television, which in its current form, can be viewed as a socially isolating medium,

although paradoxically broadcasts are essentially a shared media involving millions of people at the same time. Extending the experience of this media could imply modifying our interface to television into one whereby the viewer relaxes in their own home complete with virtual friends and family occupying virtual chairs all watching (and sharing) the same telepresence experience, and perhaps also sharing in the joint excitement or disappointment of an interaction. Imagine if, when you are on the road travelling, your hotel room allowed you to virtually join your family in a computer-generated living room — laughing at the same jokes, seeing the happiness in a child's eyes while you play a game, reading a bedtime story with expressive voices and faces, all the time allowing awareness of each other's feelings — facilitating the kind of full-bandwidth communication that brings people closer, in spite of physical distance.

3. Background — basics of emotion

Nearly a hundred definitions of emotion have been collected from the literature [15], and will not be repeated here. Internal human states like fear, anger, sadness, and joy, which psychologists call 'affective states', are examples of emotion upon which there is wide agreement. But what about interest, boredom, hope, frustration, and lust? Emotion theorists do not agree on precisely what is and is not an emotion. Fortunately, this lack of a solid definition does not mean that we cannot proceed to conduct solid research. To quote John McCarthy: 'We can't define Mount Everest precisely — whether or not a particular rock or piece of ice is or is not part of it; but it is true, without qualification, that Edmund Hillary and Tenzing Norgay climbed it in 1953. In other words, we can base solid facts and knowledge on structures that are themselves imprecisely defined.' Most people's intuitive concept of emotion is well-established, and serves the purposes of this paper well, for beginning to understand how a broad range of affective phenomena significantly influence mixed media telepresence.

Many theorists have suggested that there are 'basic emotions' and have defined these to satisfy different criteria — for example, to have a basic facial expression, such as a joyful smile. However, by this definition, 'love' is not a basic emotion because it does not have a basic facial expression. Alternatively, some researchers emphasise not the physical aspects of emotions, but the cognitive aspects. A person might cognitively appraise a situation as favourable to their goals, and thereby have a joyful emotion. To recognize such an event, a computer would need to be able to perceive a situation and reason about the emotions that it might generate. Additionally, computers would benefit from being able to perceive what emotions are actually expressed in a situation. Toward this goal, there is much research describing the physical changes in a person which accompany an emotion, such as a furrowed brow when confused, raised vocal pitch when annoyed, increased

heart rate and temperature when angry, and so forth. We believe that computers need to understand both the physical and cognitive aspects of emotion.

Computers might use many representations for emotion — not just discrete categories as implied by lists of emotions such as fear, anger, joy, and sadness, but also continuously dimensioned spaces. The two most common dimensions are ‘valence’ and ‘arousal’. In general, two dimensions cannot be used to distinguish all emotions; for example, intense fear and anger lie in the same region of high arousal and negative valence. However, these two dimensions do account for many important affective phenomena (see Fig 3). Depending on the application, a representation using continuous dimensions or discrete categories may be preferred. For example, the dimension of arousal is known to be highly correlated with attention and memory [6]. If an application involves describing viewer interest, as for assessing TV broadcasts, then the dimensions of arousal and valence are very useful.

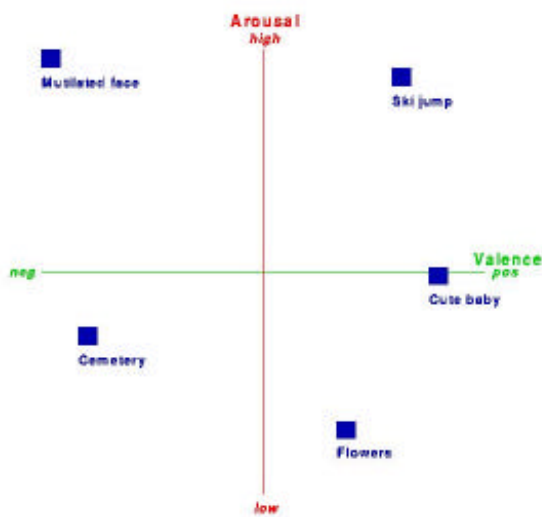


Fig 3 The dimensions of ‘valence’ and ‘arousal’ [16].

The details of what is and is not an emotion are not as important for affective intelligence as are the abilities to accurately communicate a broad range of affective phenomena. The communication of emotion, even if subtle, can powerfully influence actions. An example of the impact of emotional expression occurs in the film *Schindler's List* when a one-armed Jewish machinist goes to thank Schindler for giving him a job. His face lights up when he sees Schindler and he leans towards him with dignity and gratitude: ‘I want to thank you, sir, for giving me the opportunity to work ... the SS beat me up, they would have killed me. But, I’m essential to the war effort thanks to you ... God bless you, sir. You are a good man ... God bless you.’ The sincere joyful appreciation on the man’s face clearly affects Schindler’s feelings, and those of many

viewers. The emotions communicated in this scene become a turning point in the film, after which Schindler’s actions begin to change.

Even without trying to communicate emotion, emotions can be contagious. People may recall a favourite teacher who was so enthusiastic about a subject that they became interested in it too. Alternatively, you may have watched an actor express emotion so effectively that its impact, not only on the other characters in the drama, but also on the audience, was electrifying. In order to achieve this heightened level of communication, actors study posture, gestures, facial expressions, and vocal intonation and use a wide range of these attributes to communicate their character’s feelings. An effective actor is a master of affect.

Emotions can be communicated in many ways. The most visible form is through facial expressions. However, this form is also the easiest to control, and therefore can be the least sincere. Vocal inflection is less easy to control, and is perhaps the second most common way to communicate emotion. Other visible forms include gesture, posture, behaviour, and more subtle cues such as pupillary dilation. Pupils tend to dilate with interest and with mental workload, as well as with decreased illumination. Less visible bodily changes also accompany emotional responses — changes in galvanic skin conductivity, heart rate, blood pressure, skin temperature, respiration, muscle tension, and so on. These changes may not be noticed in casual social interaction, although some of these can be picked up from a handshake or from sitting close to a person. All of these cues, even very subtle ones, can be used to communicate important information, such as whether or not a person liked the suggestion you just made in your virtual meeting.

4. Example — the affective mirror

Imagine the most important interview of your life, with the head of the company that you have always wanted to work for. You are asked tough questions about problems you have solved, challenges you have faced, and why you want to leave your present job. At the end of this gruelling meeting, he tells you that you were too nervous-sounding, had unusually short pauses in your speech, were evasive with eye contact, and had cold clammy hands. This was not the real thing, fortunately, but a practice session in front of your trusted computer. Your computer interviewing agent, displaying the face of the chief executive, asked you questions while listening to changes in your voice and discourse parameters. It watched your facial expressions and body language, sensing changes in physiological parameters such as your skin conductivity and temperature. It watched your affective responses to see where they differed from what it usually senses from you in day-to-day interaction.

This scenario is still in the future, but most pieces of it are present technology. Inspired by the high school boy

practicing asking a girl out for a date, rehearsing in the mirror, the ‘affective mirror’ would be an agent that interacts with a person, helping individuals to see how they appear to others in various situations. With a camera, microphone, and various sensors, the system is far more advanced than the mirror most people practice in front of, even though it is potentially not as good as practising in front of a skilled human listener. Nonetheless, there are many times when a computer’s availability, patience, and non-partisan judgement cannot be beaten. Being able to ‘try out’ that important talk in front of your computer has a certain convenience and privacy that some find encouraging and comfortable. This use of affect recognition is one of many ways in which the research described here might be used in a virtual environment, not just for advancing one’s own interactive skills, but, in general, for improving communication that is mediated by technology.

5. Developing technology that communicates emotion

Present technology tends to block the transmission of affective information. We think of e-mail as having less affective bandwidth than a phone call, a phone call as having less affective bandwidth than a videoconference, and a videoconference as having less affective bandwidth than physically ‘being there.’ One of the goals of telepresence is to provide a much closer sense of being there — especially the subtle and significant aspects of emotion communication.

How might tomorrow’s enhanced reality systems facilitate truly natural and intuitive human communication? Research on developing basic techniques to assess perceived performance employs models of the human senses in order to include aspects of perception in engineering metrics. Models of hearing are used as a basis for assessing speech quality performance [17, 18], while further research is combining audio and video models for a multi-sensory assessment [19, 20]. Such an approach represents an important breakthrough since it allows performance assessment to be based on only those aspects of signal streams which are available to human consciousness. In the future such models may be embedded in the delivery technology, exploiting algorithmic models of perception to dynamically maximise subjective performance. To date, however, the emphasis in these research areas has been predominantly on tasks such as objective prediction of perceived quality, and performance optimisation in view of human sensory performance. Affective cues, such as intonation in speech, are usually treated as noise. Sometimes it is not what is said that is most important, but how it is said. The expression on a speaker’s face, the way they are leaning forward, and the subtle intonation changes in their speech can communicate powerful messages — sometimes even the opposite of what is spoken.

Tools that enable affective communication are useful not just in enhanced reality systems, but also for helping

people with various kinds of disabilities. Stephen Hawking, who relies on text-to-speech synthesis, is one of millions who is without effective natural speech communication. One of the problems with present text-to-speech systems is that they say everything with the same tone of voice. This makes it particularly difficult to communicate feelings — to interrupt angrily, to express anxious concern, to soften your voice in approval, or to indicate empathy and other expressions which illustrate a speaker’s emotional intelligence. Research is being conducted by Murray at Dundee University in Scotland [21], Alm, Murray, Arnott, and Newell [22], and by Cahn at the MIT Media Laboratory in Cambridge [23] to determine parameters of speech which these systems can modify to enable the computer to speak with suitable intonation. However, there is an interface problem, in that people talk much faster than they can type. Speech is typically around 180–250 words per minute (wpm), while a good typist may type 50–60 wpm or a stenographer with a chording keyboard may type 180 wpm. Some disabled people using standard keyboards persevere at rates of under 10 wpm. People are already limited by how quickly they can give input to a text-to-speech device. To add affective parameters suggests more control knobs for them to set, which is undesirable given that they are already busy typing as fast as possible. In some cases, when the emotion is more important than the semantics, a person might opt for hitting a button producing ‘an angry interrupting sound’ or some other ‘audio emoticon’ that conveys the emotion of the response without words. Computer recognition of affect could help in a more direct way — by recognising directly from a person’s physiological changes how their affective state is changing, and how using this can potentially help speaking disabled persons to be more expressive, without increasing their typing burden.

Within the telepresence experience that will deliver truly natural communication, we see a vision of affective audiovisual transmission. Not only should facial, gestural, and verbal communication deliver ‘affective bits’, but the system could potentially be intelligent about understanding these bits, and helping people express them, as in the affective mirror scenario above.

6. Research on affect communication

Research in computer recognition and expression of emotion is in its infancy. Current research efforts at BTL and the MIT Media Lab focus on recognition of facial expression, voice affect synthesis, and recognition of emotions from changes in physiological state parameters. Note that you cannot directly recognise somebody’s emotions — you can only recognise affective expressions and try to infer an underlying emotional state. Where this paper refers to ‘emotion recognition’, this should be interpreted as ‘combining recognition of expressions with reasoning about emotion generation to infer an underlying emotional state.’

Computers, like people, can use cognitive reasoning, a form of common sense, to understand people's goals and predict their affective state when they are disrupted. One way to recognise an expression is to record facial movements, digitise the sequence, then apply pattern recognition. Recognition from a moving sequence is generally more accurate than recognition from a still image. If, for example, a person's 'neutral' expression is a pout, only deviations from the pout (captured by video as movement) will be significant for recognising affect.

Pattern recognition can utilise a variety of techniques such as analysing individual muscle actuation parameters or (more coarsely) characterising an overall facial-movement pattern. In a test at the MIT Media Lab involving eight people, recognition rates were as high as 98 per cent for facial expressions of four emotions (see Fig 4) [24]. The emotions shown in this example are clearly exaggerated, or simulated emotions. This is analogous to controlled enunciated 'yes' and 'no' in speech recognition but clearly demonstrates how this technique can develop. Studies are under way to determine how the recognition rate changes when there are more experimental subjects. As yet, this method of recognition does not work in real time; it takes a few seconds to recognise each expression. However, advances in hardware and pattern recognition should make recognition essentially instantaneous in the near future at least for familiar expressions.

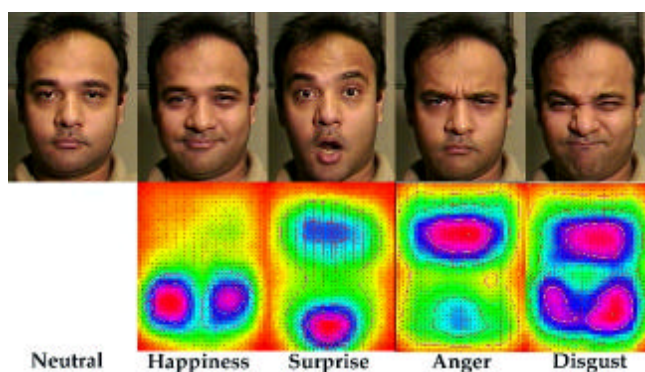


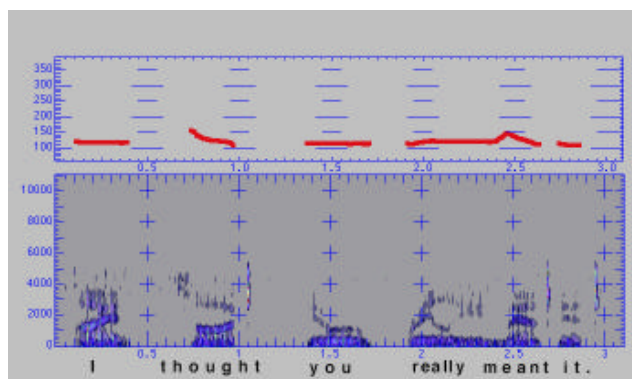
Fig 4 Facial expression recognition and colour motion energy maps. Top row: shots of the neutral expression and four others — happiness, surprise anger and disgust. Bottom row: templates of colour coded energy of each facial movement, in contrast to the still image of the neutral expression. [Courtesy of Irfan Essa]

Although facial features are one of the most visible signs of underlying emotional states, and can of course be transmitted with conventional videoconferencing techniques, they are also easy to control in order to hide emotion and can be masked with video coding techniques. One strand of our research is looking into how we perceive these hidden cues in the straight face-to-face interaction.

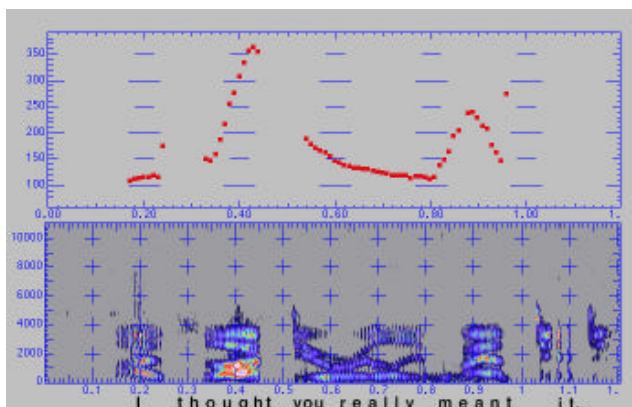
Emotional expression is clearly not limited to facial movement. Vocal intonation is the other most common way to communicate strong feelings. Several features of speech

are modulated by emotion; they are divided into such categories as voice quality (e.g. breathy or resonant, depending on individual vocal tract and breathing), timing of utterance (e.g. speaking faster for fear, slower for disgust), and utterance pitch contour (e.g. showing greater frequency range and abruptness for annoyance, a smaller range and downward contours for sadness), as illustrated in Fig 5. As these features vary, the emotional expression of the voice changes. The problem of precisely how to vary the vocal features to synthesise realistic intonation remains part of the research programme.

Preliminary experiments indicate that computers can synthesise affect sufficiently well for human listeners to recognise six different categories of emotional expression with 50-90% accuracy, significantly higher than the chance levels of 17% [23]. Humans, on average, can recognise affect with about 60% reliability from other humans [25] when tested on neutral speech or on speech where the meaning has been obscured.



(a)



(b)

Fig 5 Voice inflection synthesis. The same sentence ("I thought you really meant it") is spoken with (a) sad effect and (b) annoyed effect. The pitch track (top) and spectrogram (bottom) for each case are shown. It should be noted that the pitch range for the annoyed affect is greater than the relatively compressed range for the sad affect. The spectrograms also record differences in the speed, hesitation, and enunciation of the two cases. [Courtesy of Janet Cahn and Lisa Stifelman]

The inverse problem — computer intonation analysis, or recognising how something is said — is quite difficult. Research to date has limited the speaker to a small number of sentences, and the results are still closely dependent on the particular words spoken. A method of precisely separating what is said from how it is said has not yet been developed.

New research has also begun at the MIT Media Lab in recognising affect from physiological changes in a person, especially from signals which can easily be sensed by off-the-shelf sensors, such as a comfortable elastic heart monitor, commonly worn during athletic exercise. Four such signals are shown in Fig 6, measured from an actress while she consciously expressed anger (left) and grief (right). From top to bottom in Fig 6, the physiological signals are:

- electromyogram (microvolts),
- blood volume pulse (per cent reflectance),
- galvanic skin conductivity (microSiemens),
- respiration (per cent maximum expansion).

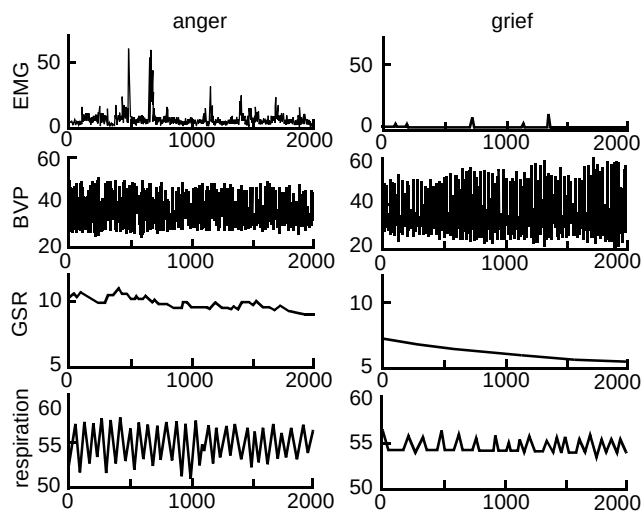


Fig 6 Physiological signals.

All of these signals can be gathered from sensors on the surface of the skin, without any pain or discomfort to the person. Each box shows 100 seconds of response. Preliminary pattern recognition results, run on various features of these signals, indicates significant recognition rates of five emotion categories, at rates 2—3 times higher than chance [26].

There are several other efforts under way, both at BTL and MIT Media Lab, to teach computers how to recognise and intelligently respond to a broad range of affective expressions — including frustration, like versus dislike,

confusion, and distress. These kinds of emotions are naturally expressed in both human/human interaction mediated by technology, and in human/computer interaction. It is therefore important not to limit them in communication.

It is to be noted that no one method — whether recognition of facial expression, voice intonation, or physiological changes — is likely to produce reliable recognition of emotion. In this sense, affect recognition is similar to other recognition problems like speech recognition and lipreading. It is probable that a personalised combination taking into account both perceptual cues (e.g. from vision and audition) and cognitive analysis of the situation is most likely to succeed. These cues will undoubtedly work best when considered in context — is it a poker game, where bluffing is the norm, or a marriage proposal, where sincerity is expected?

Although emotion researchers have worked hard to uncover universal aspects of emotional behaviour, few have explored emotions from the perspective of a person who knows another person well and can often guess their affective state. The key here is 'get to know.' Working closely with a handful of people, it can be discovered how each one of them expresses — or does not express — emotions under various circumstances. By becoming familiar with several individuals, commonalities can be detected in the ways they express emotions and perhaps used to improve recognition of affect in strangers.

There is an analogy between affect recognition research and speech recognition research in terms of the kinds of progress that it is reasonable to expect. Initially, speech-recognition systems could only recognise speech when trained on an individual speaker. Similar progress in affect recognition can be expected — initially, the best results will occur when the system is trained on the individual whose affect is being recognised. As the research and technology advance, however, the systems should improve their ability to recognise person-independent affective expressions, as has been the case in speech recognition.

There are many factors that complicate emotion recognition. The generation of emotions is influenced by goals, personality, diet, attitude, expectation, perception, and culture. Rules about the social display of emotion, e.g. the inappropriateness of showing anger, also influence its expression and make it difficult for researchers to link emotional states with forms of expression. There is presently no widely accepted comprehensive theory of emotions although much has been learned. Furthermore, nobody is 100% accurate at recognising emotion. It is clear, however, that emotions can be communicated, and that even their imperfect communication is extremely important in intelligent human interaction.

Even with the limitations of today's technology we are able to see the benefits of gesture and presence. In November 1995 BTL and MIT Media Lab demonstrated the world's first transatlantic virtual conference. This consisted of two active telepresence environments connected via an ISDN network connection. The representation of the participants is seen at the opposite end of the connection as an avatar which mirrors the movements of the host. This technique, referred to here as host and ghost, benefits conventional videoconferencing in that participants of the connection have the social etiquette to which we are accustomed in real-world interactions. Figure 7 shows an example of real and virtual representations embodied within the telepresence environment. Figure 8 also shows the state-of-the-art virtual human [27].



Fig 7 Example of a virtual conference, mixing real video with avatars.



Fig 8 State-of-the-art virtual human.

7. Towards truly personal computers

Whether through a mouse, keys, joy stick, or touch screen, many people have more physical contact with computers than they do with other people. Moreover, you can now wear computers in your shoes, shirt pocket, or belt, as clothing or accessories. Wearable computers, especially when they become as common as watches, will have the opportunity to get to know their wearer in a variety of situations. They could have access not only to the signals shown above, but also to subtle changes in temperature, blood pressure, and possibly even internal biochemistry such as blood sugar levels. Instead of being restricted to perceiving only your visible and vocal forms of affect expression, they could get to know you intimately — or as well as you will permit them to. At this point, they will also, like underwear, probably cease to be shared and will become truly personal computers. The MIT Media Lab has developed a prototype affective wearable computer, which can gather the signals shown above [28]. As this work proceeds, it will be important to highlight the difference between recognising affective symptoms which are available in face-to-face situations, and those that can be found by slightly more invasive sensors, needed in this preliminary research.

Affective information could also be communicated in unconventional ways. Imagine that a child's wearable computer could detect the interest or excitement in the classroom and broadcast it to their teacher or virtual tutor. The result would be a sort of 'mood ring' that alerts the teacher to the affective (or receptive) state. Consider the same technique being applied to a boardroom presentation, or to a large audience-participation game. A real need for such communication becomes apparent if you have ever given a lecture by videoconference to a large audience. There, in particular, you usually cannot see the details of the faces and posture of the people in the audience, whether they are expressing confusion or interest, enthusiasm or boredom. A barometer which portrays audience arousal would be useful feedback for the remote speaker, so they could adjust their presentation more effectively to the audience's response — changing their speaking style and telling more stories or jokes if they are falling asleep, or maintaining the present pace if the audience is eagerly listening. This information from the audience might be detected by sensors, which are wearable or embedded into things that they are naturally in physical contact with, such as the arm of their chair, their shoes or eyeglasses, or a ring or writing instrument.

Applications of affective recognition could extend to entertainment as well; for example, interactive games might detect a player's level of fear and give bonus points for courage. When the responses of a student playing the

computer game Doom were measured, the electromyogram of jaw clenching was expected to peak during high-action events such as when a new deadly enemy starts an attack. Indeed, we found many small peaks during stressful events of the game. However, we were surprised to find that the biggest peak — significantly higher than all the others — occurred not during a point of game action, but occurred when the student had trouble with the software (see Fig 9).

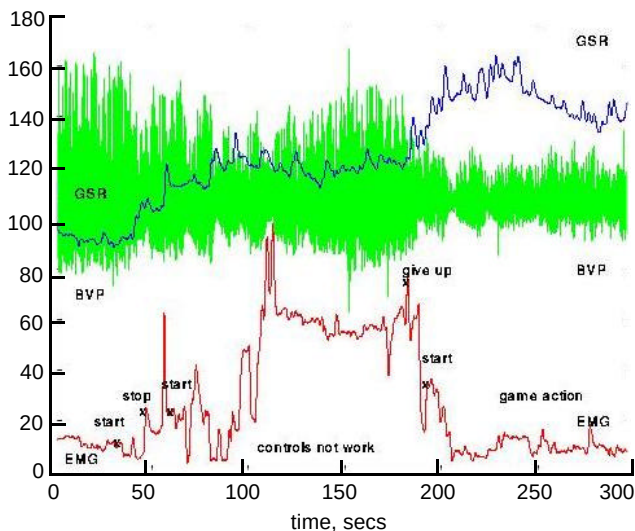


Fig 9 'Doom' electromyogram.

Computers coupled with suitable sensors and pattern-recognition algorithms should soon be able to recognise the basic affective states of a willing individual in a typical context. The emphasis on 'willing' participant here is important. Measurements of affective states obtained in an underhanded manner are not likely to be accurate. People who want to deceive such systems will probably succeed, and this is the right approach. For example, one subject managed to fool a polygraph (which gathers physiological signals similar to those used in affect recognition) by putting a thumbtack in his shoe under his big toe; he stepped on it every time he was questioned in a particular way. Affective computing will be most accurate, and useful, when it is willingly communicated, presumably for the mutual benefit of everyone involved. There are potentially harmful abuses of this technology, as with most technology, but safeguards can be evolved to prevent most of these from occurring [26]. The effect of mood on perception shows that facial expression as a communicator of affect is not a simple process. The possibilities for positive and negative feedback between affect perception and expression must be very significant — maybe our whole feeling about a conversation can be determined by a few seconds of affect exchange, like modems talking to each other.

There are many other applications of affect recognition for making smarter technology. Much research on software agents focuses on giving them the ability to learn their user's preferences. The way a human friend learns your

preferences is largely by passively watching what you do and do not like. Computers could potentially be given this ability, so that they could learn to adjust their own behaviour in accord with your affective expressions of like and dislike. A simple example is learning when to interrupt you. Intelligent people do not interrupt you willy-nilly, especially if you look annoyed when they barge in. If they repeated their interruptions despite the disapproving glances you gave, then you would consider them to be unintelligent, to say the least. In contrast, the telephone and computer interrupt us all the time, but do not have the ability to see if we were pleased by the interruption or angered. These devices, like agents, could learn to adjust their behaviour based on the cues that humans naturally use for communicating preferences — affective expressions. One of the key pieces they are missing, however, is the ability to recognise human affective expression.

8. Where do we go from here?

"Sometimes the truth of a thing is not so much in the think of it, but in the feel of it."

P Stanley Kubrick, 2001: *Filming the Future*

Scientific evidence indicates that emotions are not a luxury; they are essential for 'reason' to function normally, even in rational decision-making, and they are a natural and significant part of human interaction and face-to-face communication. Whether it is used to indicate like/dislike, interest/disinterest, or excitement/disappointment, emotion plays a key role in human communication. Affective intelligence is a new area of research which will permit technology-mediated communication to proceed more naturally among people. By giving computers the ability to recognise affect, and by considering how this can contribute to more natural human/computer interfaces and telepresence applications, the whole nature of the interface can be changed. As technology matures and the focus shifts toward human-centred issues, it is absolutely essential to pay more careful attention to the affective responses of people — the customers of the new technology. Consider, for example, a personal numbering system which allowed the affective state of the caller to anonymously become a part of the call routing process, taking into account such issues as the potential impact on the caller of routing their call to a voice messaging service rather than direct to the intended recipient.

Several potential applications to telepresence environments have been described — an affective mirror for virtual role-playing, a barometer for videoconferenced lectures, affective communication in shared spaces so that avatars react in an appropriate manner, and virtual presence games that sense affective responses. Additional applications for computers that can interpret and understand affect include better learning systems (computer recognises interest, frustration, or pleasure of pupil/subject or inhabitant), and even wearables that sense chronic anger or distress, helping

provide personalised feedback for those interested in reducing stress, to improve their overall health.

In many ways, research in affective intelligence is basic to developing understanding about humans. Human emotion theory has been held back in many ways by the inability to run experiments in natural and social situations, outside of artificial laboratory environments. By developing pattern-recognition and signal-processing tools for affective information, it is hoped to make a substantial contribution to research on basic human emotions. This is a key element of understanding the process of communications between people and the factors that influence both the communications process and the perception of that process by the participants. Furthermore, by developing such an understanding it should be possible to make better informed decisions regarding the styles or forms of communication which will be in demand in the future, and thence the nature of the technologies needed to underpin these new forms of communication.

By giving computers affective intelligence, we give them the ability to sense the affective responses of people, to help people in communicating affect, and to begin to intelligently adapt their behaviour to people. The ultimate goal is one of human-centered technology, where computers adapt to people and facilitate what comes naturally to people, instead of vice versa. Telepresence environments in particular need to facilitate the kind of interaction that comes naturally to humans. Emotions should be allowed to be communicated seamlessly, and subtly, as is natural among people sharing the same physical space.

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